Q1) a) here is snapshot from the fedora :

```
/usr/bin/checkmodule:  error(s) encountered while parsing configuration
make: *** [/usr/share/selinux/devel/include/Makefile:157: tmp/FaishalYoushuf.mod] Error 1
root@fedora:~# nano /root/FaishalYoushuf.te
root@fedora:~# make -f /usr/share/selinux/devel/Makefile
Compiling targeted FaishalYoushuf module
Creating targeted FaishalYoushuf.pp policy package
rm tmp/FaishalYoushuf.mod.fc tmp/FaishalYoushuf.mod
root@fedora:~# semodule -i FaishalYoushuf.pp
root@fedora:~# restorecon -Rv /home/Adm_FaishalYoushuf
root@fedora:~# exit
logout
[1]+  Done                    ./FaishalYoushuf_app
Adm_FaishalYoushuf@fedora:~$ cd /home/Adm_FaishalYoushuf
./FaishalYoushuf_app &
ps -eZ | grep FaishalYoushuf
[1] 14239
line1
line2
```

b) for 1b)

```
Mar 29  3:48 AM
faishalyoushuf@fedora:~/Downloads — -bash
~/Downloads
faishalyoushuf@fedora:~$ sudo useradd -m Adm_FaishalYoushuf
sudo passwd Adm_FaishalYoushuf
[sudo] password for faishalyoushuf:
useradd: user 'Adm_FaishalYoushuf' already exists
New password:
Retype new password:
passwd: password updated successfully
faishalyoushuf@fedora:~$ targetprogram.c
bash: targetprogram.c: command not found...
faishalyoushuf@fedora:~$ cd ~/Downloads
ls
targetprogram.c
faishalyoushuf@fedora:~/Downloads$ sudo cp targetprogram.c /home/Adm_FaishalYoushuf/
faishalyoushuf@fedora:~/Downloads$ su - Adm_FaishalYoushuf
Password:
Adm_FaishalYoushuf@fedora:~$ gcc -o FaishalYoushuf_app targetprogram.c
Adm_FaishalYoushuf@fedora:~$ mkdir -p ~/priv
echo "line1" > ~/priv/params.txt
echo "line2" >> ~/priv/params.txt
touch ~/priv/records.txt
Adm_FaishalYoushuf@fedora:~$ ./FaishalYoushuf_app
line1
line2
You have a large params file, that's good to start with for testing...
su -Adm_FaishalYoushuf@fedora:~$ su -
Password:
root@fedora:~# getenforce
Enforcing
root@fedora:~# semanage fcontext -a -t FaishalYoushuf_exec_t "/home/Adm_FaishalYoushuf/FaishalYoushuf_app"
restorecon -v /home/Adm_FaishalYoushuf/FaishalYoushuf_app
ValueError: Type FaishalYoushuf_exec_t is invalid, must be a file or device type
root@fedora:~# su -
root@fedora:~# sepolicy generate --application -n FaishalYoushuf /home/Adm_FaishalYoushuf/FaishalYoushuf_app
bash: sepolicy: command not found...
Install package 'policycoreutils-devel' to provide command 'sepolicy'? [N/y] y
```

c) for 1c)

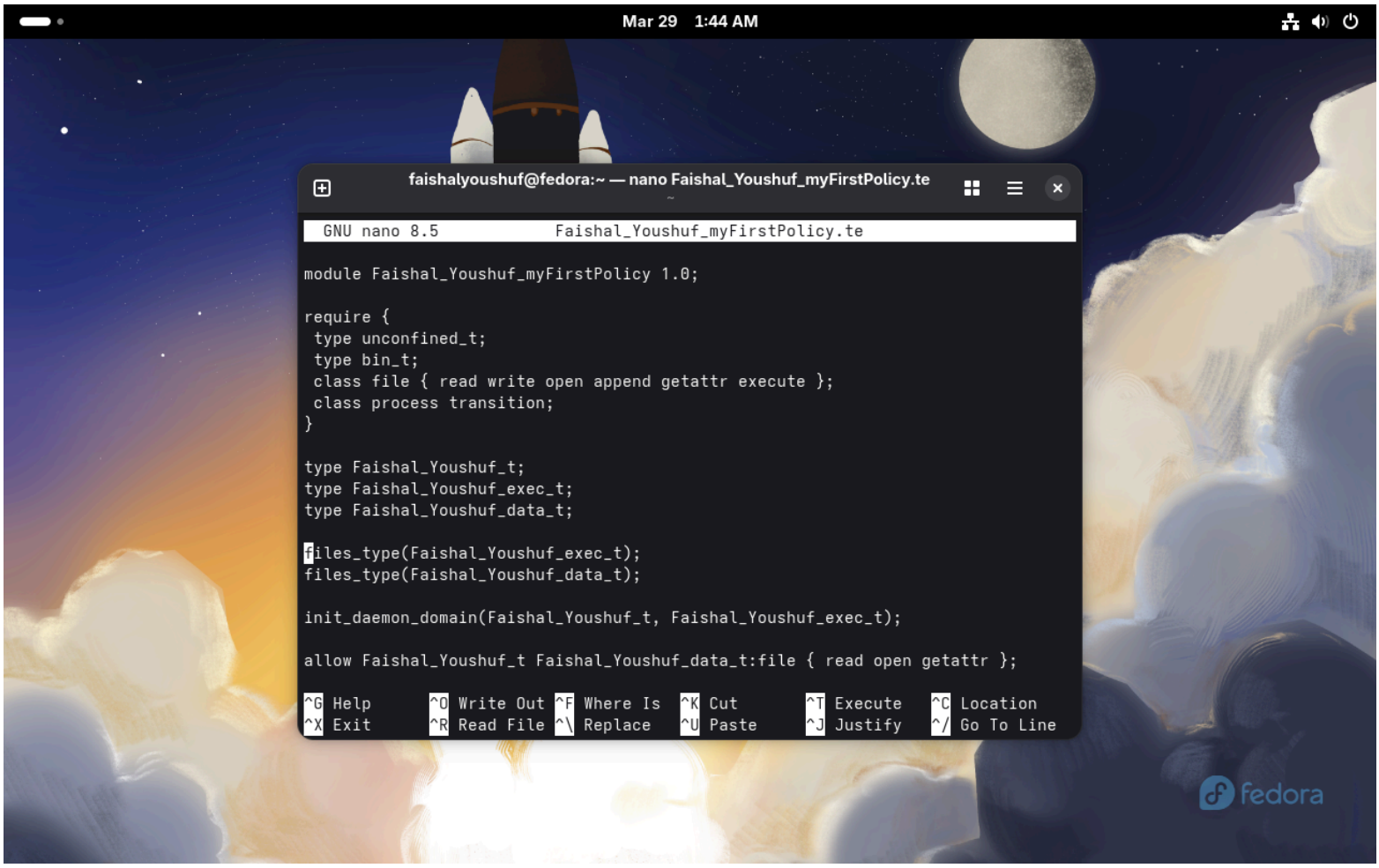
Mar 29 3:50 AM

faishalyoushuf@fedora:~/Downloads — -bash
~/Downloads

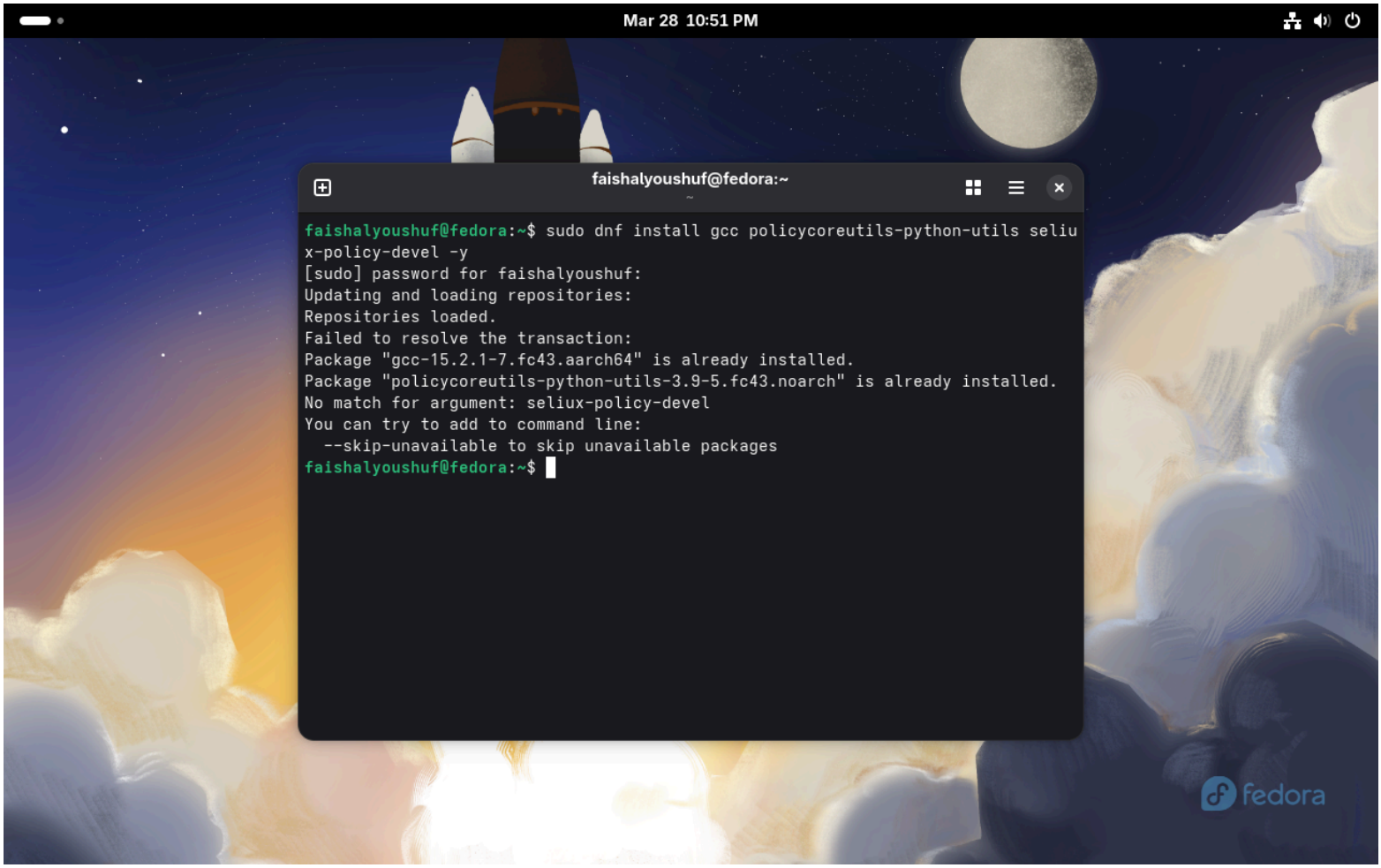
```
You have a large params file, that's good to start with for testing...
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 13303 pts/0 00:00:00 FaishalYoushuf_
Adm_FaishalYoushuf@fedora:~$ su -
Password:
root@fedora:~# nano /root/FaishalYoushuf.te
root@fedora:~# make -f /usr/share/selinux/devel/Makefile
Compiling targeted FaishalYoushuf module
FaishalYoushuf.te:15:ERROR 'Duplicate declaration of type' at token ';' on line 3622:
type FaishalYoushuf_exec_t;

/usr/bin/checkmodule:  error(s) encountered while parsing configuration
make: *** [/usr/share/selinux/devel/include/Makefile:157: tmp/FaishalYoushuf.mod] Error 1
root@fedora:~# nano /root/FaishalYoushuf.te
root@fedora:~# make -f /usr/share/selinux/devel/Makefile
Compiling targeted FaishalYoushuf module
Creating targeted FaishalYoushuf.pp policy package
rm tmp/FaishalYoushuf.mod.fc tmp/FaishalYoushuf.mod
root@fedora:~# semodule -i FaishalYoushuf.pp
root@fedora:~# restorecon -Rv /home/Adm_FaishalYoushuf
root@fedora:~# exit
logout
[1]+  Done                       ./FaishalYoushuf_app
Adm_FaishalYoushuf@fedora:~$ cd /home/Adm_FaishalYoushuf
./FaishalYoushuf_app &
ps -eZ | grep FaishalYoushuf
[1] 14239
line1
line2
You have a large params file, that's good to start with for testing...
unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023 14239 pts/0 00:00:00 FaishalYoushuf_
Adm_FaishalYoushuf@fedora:~$ ls -Z FaishalYoushuf_app
ps -eZ | grep FaishalYoushuf
unconfined_u:object_r:FaishalYoushuf_exec_t:s0 FaishalYoushuf_app
[1]+  Done                       ./FaishalYoushuf_app
Adm_FaishalYoushuf@fedora:~$
```

Here I provided the .te file I used for policy writing :



Some of other snippets of my fedora :



2) module FaishalYoushuf myFirstPolicy 1.0;

```
require {
    type unconfined_t;
    type user_home_t;
    class file open getattr write execute append create read getattr write create execute append create write getattr write create append execute getattr write create execute create getattr write create execute append create write getattr write create execute create getattr write create execute create create getattr write execute create getattr write execute create create getattr write execute create getattr write create execute create create getattr write execute create getattr write create execute create getattr write execute create getattr write create execute create getattr write execute create getattr write create execute create getattr write execute create getattr write create execute create getattr write execute create getattr write create execute create getattr write create
    class dir { search };
}

/* Define process domain */
type FaishalYoushuf_t;

/*
```

```
Explanation:
- specifies a new SELinux domain of the application.
- This locks down the application such that it does not execute in unconfined t.
*/

/* Define executable type */
type FaishalYoushuf_exec_t;

/*
Explanation:
- Attaches labels on the application binary.
- Necessary to monitor which file activates the new domain by SELinux.
*/

/* Allow domain transition */
domain_type(FaishalYoushuf_t)
domain_entry_file(FaishalYoushuf_t, FaishalYoushuf_exec_t)

/*
Explanation:
- domain as FaishalYoushuf t:Makes FaishalYoushuf t a valid process domain.
- domain entry file: It enables running of a binary which switches between user domain to
FaishalYoushuf t domain.
*/

/* permit user domain execution.
Permission unconfinedFaishalYoushufExecutet:file execute;

/*
Explanation:
- Permits the user (unconfined_t) to run the application.
- Prerequisite to the application to start and switch domains.
*/

/* Allow reading params.txt */
add FaishalYoushuf_t user_home_t:file getattr open read;

/*
Explanation:
- The program reads priv/params.txt.
- Can read, open and getattr permission to file contents.
```

*/

/* Allow writing to records.txt */
user hometrees FaishalYoushuf_t:write user hometree open getattr create file write append open getattr create user hometree write append open getattr create file write append open getattr create user hometree write append open getattr create file write append open getattr create user hometree write append open getattr create file write append open getattr create user hometree write append open getattr create file write append open getattr create user hometree write append open getattr create file write append

/*
Explanation:
- The application appends/writes to the file records.txt/priv.
- append allows adding data.
- write allows modification.
- create: the file can be created in the case it is absent.
*/

/* Allow directory access */
user-dir-search- FaishalYoushuft user-home-t;

/*
Explanation:
- Permitted to write to files in folder priv.
- SELinux needs to have permission of directory searching.
*/

/* Permit the running of /usr/bin/strings.
user This also executes a file called FaishalYousef_t user_home_t.

/*
Explanation:
- Small file in the form of parameters causes execution of /usr/bin/strings.
- Permission is executed by needs over that binary.
*/

Short explanation for this policy by me :

The policy uses least privilege and permits only the permissions necessary to make the application run. The application deserts a parameter file, outputs to a records file, loads a system binary, and accesses a directory. All the rules are related to such actions directly, and no unneeded permissions are provided.